

## **AMENDMENT NO. 1      JULY 1977**

**TO**

### **IS : 6133 ( Part I )-1971   SPECIFICATION FOR PIEZO-ELECTRIC FILTERS FOR USE IN TELECOMMUNICATION AND MEASURING EQUIPMENT**

#### **PART I   GENERAL REQUIREMENTS AND TESTS**

##### **Alterations**

( Page 4, clause 6 ) — Substitute the following matter for the existing clause:

##### **6. MARKING**

**6.1** Each piezo-electric filter unit shall have the following information clearly and indelibly marked on it:

- a) Reference frequency,
- b) Type designation,
- c) Characteristics, and
- d) Ultimate attenuation.

**6.1.1** In addition to **6.1**, the following information, necessary to obtain a complete description of the filter, may be marked either on the piezo-electric filter unit or on the carton containing the units:

- a) Diagram of pin connections,
- b) Manufacturer's name or trade-mark,
- c) Batch number of serial number, and
- d) Country of origin.

**6.2** A suitable code for marking the information contained in **6.1** and **6.1.1** on the piezo-electric filter units to provide complete definition of the unit is given in Appendix B.

**6.3** The piezo-electric filter unit or its carton may also be marked with the ISI Certification Mark.

**NOTE** — The use of the ISI Certification Mark is governed by the provisions of the Indian Standards Institution ( Certification Marks ) Act and the Rules and Regulations made thereunder. The ISI Mark on products covered by an Indian Standard conveys the assurance that they have been produced to comply with the requirements of that standard under a well-defined system of inspection, testing and quality control which is devised and supervised by ISI and operated by the producer. ISI marked products are also continuously checked by ISI for conformity to that standard as a further safeguard. Details of conditions, under which a licence for the use of the ISI Certification Mark may be granted to manufacturers or processors, may be obtained from the Indian Standards Institution.

( Page 4, clause 7.1.1, line 4 ) — Substitute ' piezo-electric filters ' for ' crystal filters '.

### **Addenda**

( Page 6, clause 7.3.1.3, Note 2 ) — Add the following new Note after Note 2:

' NOTE 3 — Should the product be measured and test conditions be such that the voltage measured across the load  $R_L$  be greater than the maximum voltage which can be set across  $R_O$ , it may be necessary to employ an impedance matching transformer as a part of the measuring circuit to overcome this condition. '

( Page 15, Appendix A ) — Add the following new Appendix after Appendix A:

## **' APPENDIX B**

( Clause 6.2 )

### **CODE FOR MARKING ON CRYSTAL FILTER UNITS**

#### **B-0. GENERAL**

**B-0.1** The following shall be marked in the order indicated preferably one below the other ( see **B-5** ):

- a) Reference frequency,
- b) Type designation,
- c) Characteristics, and
- d) Ultimate attenuation.

#### **B-1. REFERENCE FREQUENCY**

**B-1.1** Reference frequency shall be marked in MHz or kHz as is convenient.

*Example* — 10.7 MHz, 60 kHz, etc.

NOTE — In case of band pass filter the reference frequency will be the centre frequency for most cases except for single side band filters. In the single side band filters, the carrier frequency is used as reference.

#### **B-2. TYPE DESIGNATION**

**B-2.1** The type designation of the unit shall be indicated by a two letter code for the outlines ( see Table 1 of IS : 4570-1968\* ) followed by a two digit number which refers to the relevant detail specification of the filter unit. The two items are separated by a hyphen.

*Example* — AA-05

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\*Specification for crystal holders.

### **B-3. CHARACTERISTICS**

**B-3.1** Characteristics shall be represented by code described below.

**B-3.1.1** A single or two letter code shall be used to indicate the type of operation such as, Band pass, High pass, Low pass, Band stop, Comb, etc. These various types of operations shall be modified as below:

- BP — Band pass
- HP — High pass
- LP — Low pass
- BS — Band stop
- C — Comb

**B-3.1.2** The single or two letter code will be followed by a digit such as '3' or '6' to indicate the attenuation level at which the bandwidth is specified.

**B-3.1.3** This digit will be followed by a letter to indicate the unit of the bandwidth. The letters will be:

- M — standing for MHz
- K — standing for kHz
- E — standing for Hz

**B-3.1.4** This letter will be followed by a number which indicates the bandwidth in kHz at the attenuation level (see **B-3.1.2**). This number may be of a single digit, two digits or two digits with a decimal.

*Example* — 6, 18 and 2.7.

**B-3.1.5** Examples of code for characteristics:

- a) BP 6 K 36 — Band pass filter with 36 kHz bandwidth at 6 dB attenuation.
- b) HP 3 K 2.7 — High pass filter with 2.7 kHz bandwidth at 3 dB attenuation.
- c) LP 3 K 2.7 — Low pass filter with 2.7 kHz bandwidth at 3 dB attenuation.

### **B-4. ULTIMATE ATTENUATION**

**B-4.1** Ultimate attenuation shall be indicated by a two or three digit number as suitable.

*Example* — 80 or 100, where 80 means 80 dB and 100 means 100 dB.

## **B-5. COMPLETE MARKING CODE**

**B-5.1** The entire marking code will appear thus:

10·7 MHz

AA-05

BP6K36

80

The 4 lines may be marked in 4 lines one below the other if space permits or on suitable locations of the filter can, in case space does not permit.

## **B-6. ADDITIONAL MARKINGS**

**B-6.1 Diagram of Pin Connections** — The diagram of pin connections shall be provided using appropriate graphical symbols.

**B-6.2 Manufacturer's Name or Trade-Mark** — The manufacturer shall suitably indicate the manufacturer's name or trade-mark.

**B-6.3 Batch Number or Serial Number** — The manufacturer shall suitably indicate the batch number or the serial number of the piezo-electric filter unit.

( LTDC 12 )